FILE HANDLING IN PYTHON

File handling allows us to read from and write to file - common task in almost every real-world application (s… data, logs, reports, etc)

Why File Handing?

- To save user data permanently (like note, records, etc)
- To read content from a file (like student marks exam re…
- Helps build real applications like!
  - Address books
  - Billing systems
  - Exam report generators

Python File modes

Mode	Meaning	Use case
'r'	Read (default mode)	Read an existing f…
'w'	write (overwrite if file exists)	write new content
'a'	Append (adds content at the end)	Add data without dele… old
'x'	Create (fails if file exists)	create new file safe…
'b'	Binary mode	For image, videos et…
't'	Text mode (default	For text files.

Opening a File

file = open ("students.txt", "r)

Parts:

- "Students.txt" → file name
- "r" → mode (read mode)

Reading From a File:

1. read() - Reads entire file

```
file = open ("notes.text", "r")
Print (file.read())
file.close()
```

(2) readline() - Read one line

```
file = open ("notes.txt", "r")
Print (file.readline())
file.close()
```

(3) readlines() - Reads all lines into a list

```
file = open ("notes.txt", "r")
lines = file.readlines()
Print (lines)
file.close()
```

write() - Overwrites entire file

```
file = open ("data.txt", "w")
file.write (" namskara Bengaluru! \n")
file.write (" Python is awesome!")
file.close()
```

Appending to a File

```
file = open("data.txt", "a")
file.write("\n This line is added later!")
```

Using with statement (Best practice)

```
with open("student.txt", "r") as file:
    content = file.read()
    print(context)
```

Writing List of Data to File

```
students = ["Ravi, "meena", "Dinesh"]
with open('students.txt', "w") as file:
    for student in students:
    file.write(student + "\n")
```

Reading File and Processing Each Line

```
with open("student.txt", "r") as file:
    for line in file:
        print("student:", line.strip())
```

- strip() removes \n from the end of each line.

Common Errors

Error	Cause
FileNotFoundError	Trying to read a missing file
Permission Error	Access denied
IsADirectory Error	Trying to open a folder as file.